

## 练习三十 几何光学（二）

1. 发散、会聚

2. 电子德布罗意波波长短

3. (2); 4. (4)

$$5. \quad \frac{1}{\infty} + \frac{1}{-20} = \frac{1}{f} \quad \phi = \frac{1}{f} = \frac{1}{-0.2} = -5 D$$

$$\frac{1}{10} + \frac{1}{-25} = \frac{1}{f} = \phi \quad \phi = 10 - 4 = 6 D$$

$$6. \quad v_1 = 22 - 2 = 20 \text{ cm}$$

$$\frac{1}{u} + \frac{1}{v_1} = \frac{1}{f_1} \quad \frac{1}{u} + \frac{1}{20} = \frac{1}{0.5} \quad u = 0.513 \text{ cm}$$

$$M \approx \frac{25s}{f_1 f_2} = \frac{25 \times 22}{0.5 \times 2} = 550 \quad \text{或} \quad M = m\alpha = \frac{v_1}{u} \cdot \frac{25}{f_2} = \frac{20}{0.513} \times \frac{25}{2} = 487.5$$